PA1: Shape-Following Robot

Xiaoming Zhao
Screen Recording: [Google Drive Link]

April 17, 2026

Implementation

Two ROS 2 Humble nodes drive a TurtleBot3 through three user-selected shapes in Gazebo. Three type of shape was implemented, including isosceles trapezoid parameterized by radius r, "D" shape with straight stroke $2r$ plus semicircle of radius r, and general closed polygon from arbitrary (x, y) vertices in the `odom` frame.

Open-loop Each shape decomposes into turn-then-drive segments. Turns use a proportional heading controller ($\omega = 1.5\, e_\theta$, clamped to ± 0.3 rad/s) with $R_{\text{ICC}} = 0$ ($v_l = -\omega L/2$, $v_r = \omega L/2$). Straight segments command constant $v = 0.15$ m/s ($v_l = v_r$, ICC at infinity) and terminate on odometry distance. The semicircle is discretized into 12 waypoints at 15° intervals, each reached by the same turn-then-drive primitive. Control runs at 10 Hz with tolerances $\delta_p = 0.05$ m, $\delta_\theta = 0.02$ rad.

Closed-loop Adding three improvement for tighten trajectory tracking, including a PID heading controller (K_p=2.0, K_i=0.02, K_d=0.2) replaces the P-only law; the integral resets below 0.05 rad to prevent oscillation. Straight segments correct cross track error every cycle $\theta_{\text{des}} = \text{atan2}(\Delta y, \Delta x) + \arctan(K_{\text{ct}} \cdot e_{\text{ct}})$, where e_{ct} is the signed perpendicular distance to the segment line and $K_{\text{ct}} = 0.8$. The polygon controller uses `tf2_ros lookupTransform` to compute goal in the robot frame. Control runs at 20 Hz with $\delta_p = 0.03$ m, $\delta_\theta = 0.02$ rad

Both nodes publish `nav_msgs/Path` on `/actual_path{_cl}` and `/ideal_path{_cl}` for RViz visualization and write per-run CSV logs (`trajectory_*.csv`, `waypoints_*.csv`) to `~/ros2_ws/logs/`

Evaluation

All tests use $r = 5$ m or similar vertex spacing. Closure error is the Euclidean distance from the robot's final position to the start point after completing the full shape.

Shape	Controller	Closure Error (m)	Improvement
Trapezoid (r=5)	Open-loop	0.1813	—
Trapezoid (r=5)	Closed-loop	0.0261	6.9×
D-shape (r=5)	Open-loop	0.0394	—
D-shape (r=5)	Closed-loop	0.0005	78.8×
Polygon (diamond, ±5)	Open-loop	0.2027	—
Polygon (diamond, ±5)	Closed-loop	0.0282	7.2×

The closed-loop controller reduces closure error by 7−79× with all shapes. The D-shape benefits most because the waypoint-discretized semicircle combined with cross-track correction on each chord segment eliminates radial drift. Straight-segment-dominant shapes show few improvement, mostly from cross-track correction eliminating cumulative lateral drift and PID damping reducing heading overshoot at vertices.

Credits

Developed by Xiaoming Zhao, referecing from lecture materials, slides, and sample codes. Debugging, PID tuning, and report formatting assisted by Generative AI, including Claude and Gemini.